Optimal Molecular Networks for Excitonic Energy Transport

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March 18, 2011

Long-lived Quantum Coherence in Photosynthesis

Does coherence benefit transport efficiency?[1,2]

coherence experimentally verified[1,2] in Fenna-Matthews-Olson (FMO) protein trimer of green sulfur bacteria

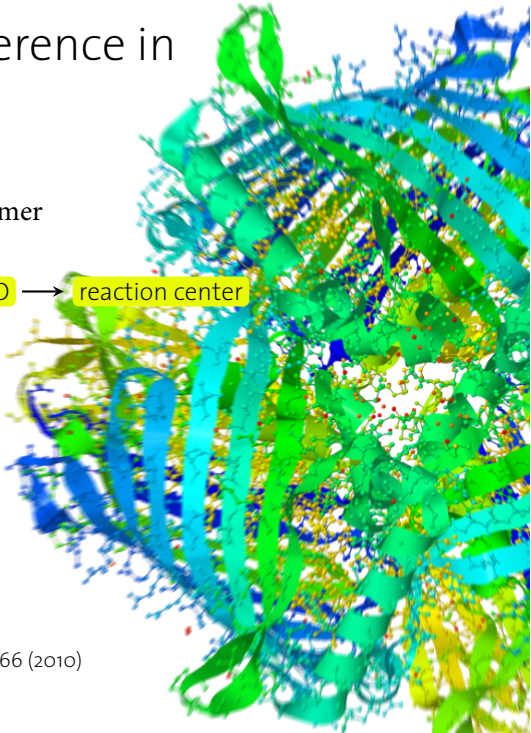
energy shuttle photon ⟶ antenna ⟶ FMO ⟶ reaction center

efficiency 95% > classical prediction

mechanism role of quantum coherence?

transport via 7 molecular sites per monomer

coherence preserved by nontrivial, noisy environment



[1] Engel et al., Nature, **446**, 782 (2007)

[2] Panitchayangkoon et al., Proc. Natl. Acad. Sci. USA, **107**, 12766 (2010)

Open Question

Which mechanisms drive FMO's efficiency?

noise enhanced

FMO Hamiltonian[1] predicts suppression of transport due to *destructive* quantum interference

noise eliminates interference, transition to *classical* transport

optimized bath optimal control of transport[2,3]

interference induced

interference sensitive to system parameters, *hallmark of coherence*

FMO Hamiltonian[1] has error bars

optimized system efficient *constructive* interference scenario, consistent with data

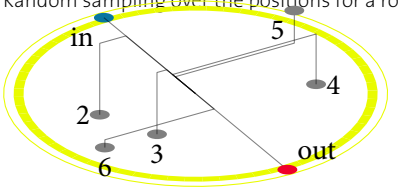
[1] Adolphs et al., Biophys. J., **91**, 2778 (2006)

[2] Rebentrost et al., J. Phys. Chem. B, **113**, 9942 (2009)

[3] Plenio et al., New J. Phys., **10**, 113019 (2008)

Robust Statistical Approach

Random sampling over the positions for a robust comparison



single excitation from ‘in’ to ‘out’

random distribution of 5 intermediates

random quantum network

$H \propto \sum_{i \neq j} r_{i,j}^{-3} \, \sigma_{i,+} \sigma_{j,-}$

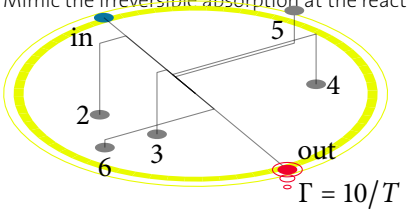
reference time $T = \frac{\pi}{2H_{\mathrm{in,out}}}$

time evolution $\dot{\rho} = -\mathrm{i}\,[H, \rho]$

timescales reference T

Robust Statistical Approach

Mimic the irreversible absorption at the reaction center



sink transfer at 'out' with rate $\Gamma = 10/T$,

drain $\mathfrak{p} \xrightarrow{t\to\infty} 1$,

average transfer time $\mathfrak{T} = \int_0^\infty t\,\dot{\mathfrak{p}}(t)\,\mathrm{d}t$

Lindbladian $L_{\mathrm{sink}}(\rho) =$

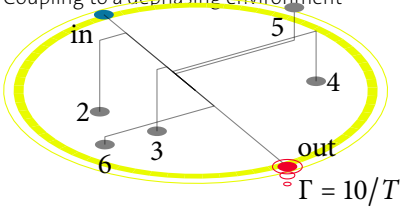
$\Gamma\left(\sigma_{\mathrm{out},-}\rho\sigma_{\mathrm{out},+} - \frac{1}{2}\{\sigma_{\mathrm{out},+}\sigma_{\mathrm{out},-}, \rho\}\right)$

time evolution $\dot{\rho} = -\mathrm{i}\,[H, \rho] + L_{\mathrm{sink}}(\rho)$

timescales reference T, sink $\Gamma^{-1} = T/10$

Robust Statistical Approach

Coupling to a dephasing environment



dephasing homogeneous rate 4γ

Lindbladian $L_{\mathrm{deph}}(\rho) = \gamma \sum_i \left(\sigma_{z,i}\rho\sigma_{z,i}^\dagger - \frac{1}{2}\left\{\sigma_{z,i}^\dagger\sigma_{z,i}\rho\right\}\right)$

time evolution $\dot{\rho} = -\mathrm{i}\,[H,\rho] + L_{\mathrm{sink}}(\rho) + L_{\mathrm{deph}}(\rho)$

timescales reference T, sink $\Gamma^{-1} = T/10$, dephasing $(4\gamma)^{-1}$

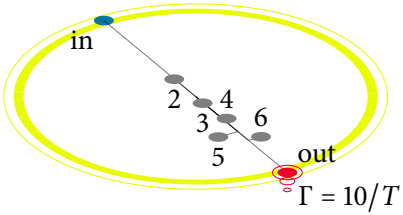
Transport without Dephasing

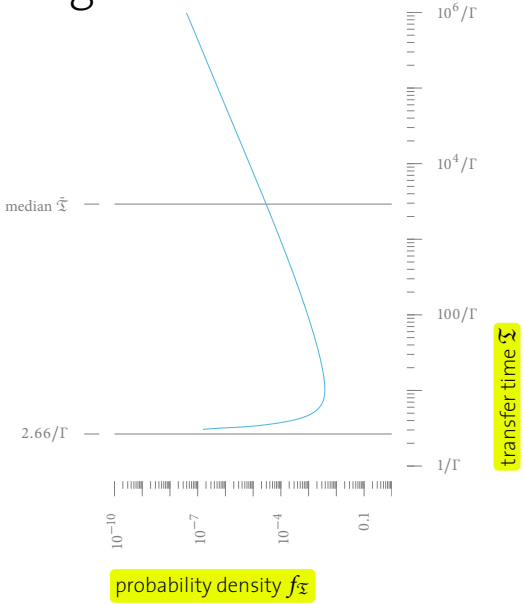
heavy-tailed pdf $f_{\mathfrak{T}} \sim_{\mathfrak{T}\to\infty} \mathfrak{T}^{-1.2}$

transfer time $\mathfrak{T}$ can be arbitrarily long
due to *destructive* quantum interference

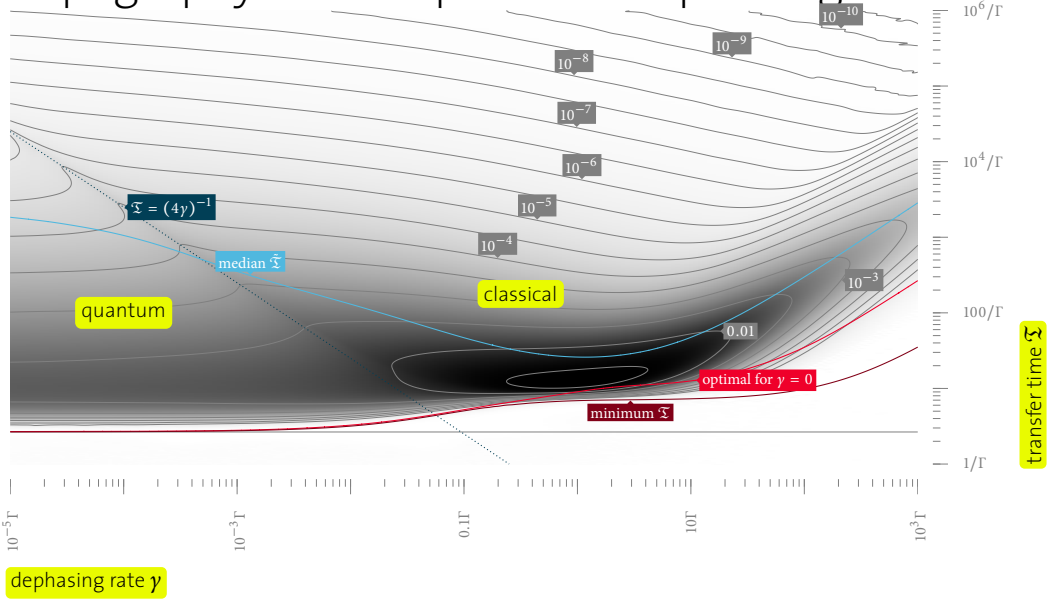
typical configuration median $\tilde{\mathfrak{T}} \simeq 292T$

optimal configuration $\mathfrak{T} \simeq 0.266T \ll \tilde{\mathfrak{T}}$
due to *constructive* quantum interference





Topography of Transport and Dephasing



Conclusions

noise enhanced

most configurations are inefficient and need noise, transport is *classical*

interference induced

rare configurations are efficient by design[2,3], transport is *coherent* and *faster* than in noise-assisted scenario[1]

[1] T. S., T. Wellens, and A. Buchleitner, arXiv:1103.2944 (2011)

[2] T. S., F. de Melo, T. Wellens, F. Mintert, and A. Buchleitner, Phys. Rev. E, **83**, 021912 (2011)

[3] T. S., F. Mintert, T. Wellens, and A. Buchleitner, Semiconductors and Semimetals, **83**, 1 (2010)

“That’s all Folks!”